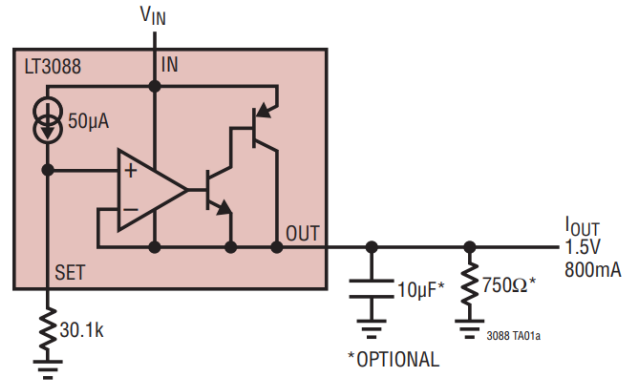


LT3088



Please walk through the datasheet **LT3088** to answer the following questions:

Google: **LT3088 datasheet**

Q1. What is the input voltage range of this regulator?

$$1.2\text{ V} < V_{in} < 36\text{ V}$$

Q2. What is the output voltage range of this regulator?

$$0\text{ V} < V_{out} < 34.5\text{ V}$$

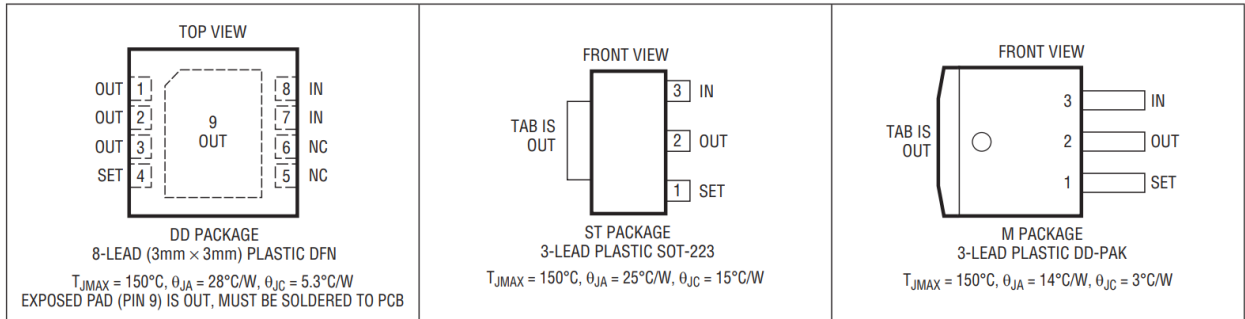
Q3. How much maximum output current this device can provide?

800mA

Q4. What is the set pin current of this regulator?

50µA

Q5. What packages are available for this regulator, and how to identify the package from the part number?



ORDER INFORMATION

LEAD FREE FINISH	TAPE AND REEL	PART MARKING*	PACKAGE DESCRIPTION	TEMPERATURE RANGE
LT3088EDD#PBF	LT3088EDD#TRPBF	LGSZ	8-Lead (3mm × 3mm) Plastic DFN	−40°C to 125°C
LT3088IDD#PBF	LT3088IDD#TRPBF	LGSZ	8-Lead (3mm × 3mm) Plastic DFN	−40°C to 125°C
LT3088HDD#PBF	LT3088HDD#TRPBF	LGSZ	8-Lead (3mm × 3mm) Plastic DFN	−40°C to 150°C
LT3088EST#PBF	LT3088EST#TRPBF	3088	3-Lead Plastic SOT-223	−40°C to 125°C
LT3088IST#PBF	LT3088IST#TRPBF	3088	3-Lead Plastic SOT-223	−40°C to 125°C
LT3088HST#PBF	LT3088HST#TRPBF	3088	3-Lead Plastic SOT-223	−40°C to 150°C
LT3088MPST#PBF	LT3088MPST#TRPBF	3088	3-Lead Plastic SOT-223	−55°C to 150°C
LT3088EM#PBF	LT3088EM#TRPBF	LT3088M	3-Lead Plastic DD-Pak	−40°C to 125°C
LT3088IM#PBF	LT3088IM#TRPBF	LT3088M	3-Lead Plastic DD-Pak	−40°C to 125°C
LT3088HM#PBF	LT3088HM#TRPBF	LT3088M	3-Lead Plastic DD-Pak	−40°C to 150°C
LT3088MPM#PBF	LT3088MPM#TRPBF	LT3088M	3-Lead Plastic DD-Pak	−55°C to 150°C

Q6. To which node of this device should we connect R_{SET} ?

Node SET.

Q7. What is a minimum load current required for the regulation of the output voltage?

2mA

Q8. Why use C_{SET} and C_{OUT} , and what are the appropriate value for these capacitors?

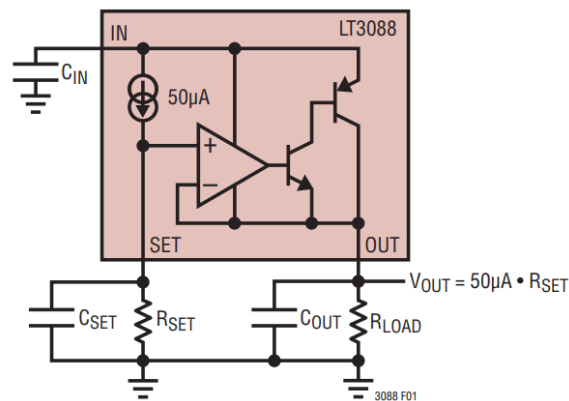


Figure 1. Basic Adjustable Regulator

C_{SET} :

Since the SET pin is a high impedance node, unwanted signals may couple into the SET pin and cause erratic behavior. The easiest way to remedy this is to bypass the SET pin a small amount of capacitance from SET to ground, 10pF to 20pF is sufficient.

This capacitor also provides a **soft-start** function for the current source.

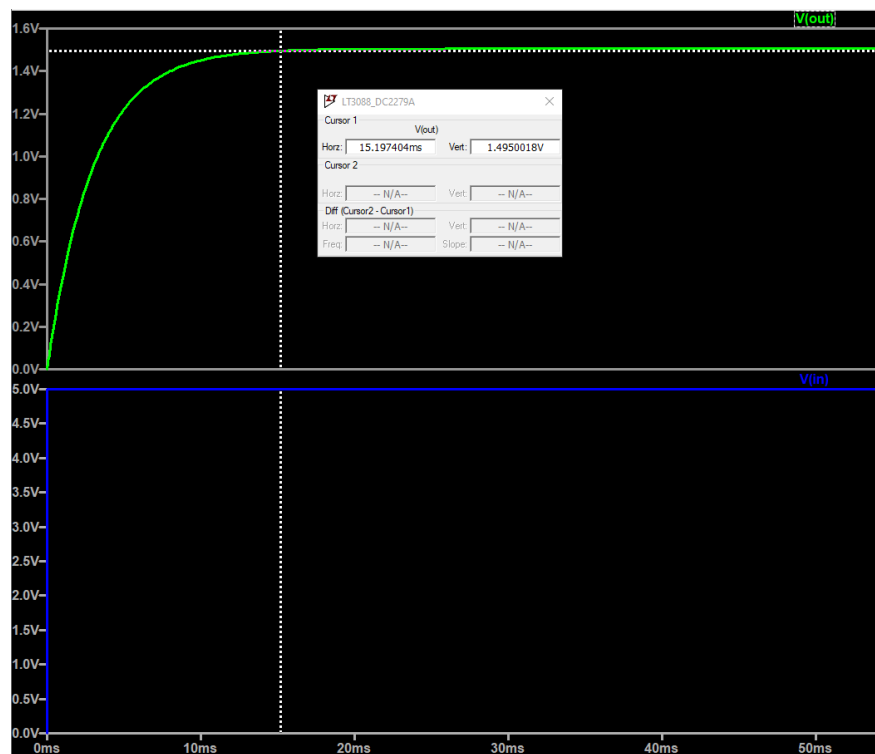
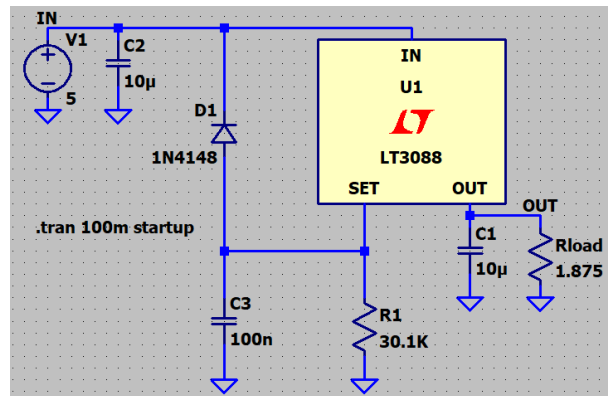
C_{OUT} :

To reject ripple at the output used C_{OUT} of about 2.2µF or 10µF.

Q9. If $R_{SET} = 30.1 \text{ k}\Omega$, calculate the output voltage. If $C_{SET} = 100\text{nF}$ is added in parallel with R_{SET} , determine the time required for the output to stabilize, and verify the result using an LTspice simulation?

$$V_{OUT} = 50\mu\text{A} \times R_{SET} = 50\mu\text{A} \times 30.1 \text{ k}\Omega = 1.5 \text{ V}$$

$$\text{Delay} = 5 \times R_{SET} \times C_{SET} = 5 \times 30.1 \text{ k}\Omega \times 100\text{nF} = 15\text{ms}$$



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